Slow Dynamics and High Variability in Clustered Networks

Reproduction of Litwin-Kumar and Doiron (2012)

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Abstract. *This report reproduces the key findings from Litwin-Kumar and Doiron (2012) on the slow dynamics and variability in clustered spiking networks. We implement their clustered network model and show that potentiated within-cluster connectivity introduces metastable firing states within the clusters. Using analyses from the original paper, we identify slow firing-rate fluctuations (via long-timescale decays in spike-count autocorrelation), higher trial-to-trial spike-count variability (via super-Poisson Fano Factors), and weak average pairwise correlations with a slightly elevated tail for within-cluster neurons. We also reproduce stimulus-quenched spiking variability. These results support the original paper's finding that clustered excitatory connections are sufficient to generate slow firing-rate fluctuations and stimulus-quenched variability in LIF networks—features also seen in cortical recordings.*

1. Introduction

Cortical networks are often modeled as unstructured, randomly connected networks with balanced excitation and inhibition. This balance can place the network in an irregular, asynchronous regime in which neurons exhibit high trial-to-trial variability in spike timing. In this regime, excitation and inhibition are large but cancel in the mean. Thus, spikes are often fluctuation-driven, resulting in high spike-time variability.

However, cortical networks also exhibit slower, population-level changes in firing rate over longer timescales (hundreds of milliseconds to seconds). Baseline, unstructured, balanced excitation-inhibition networks do not typically capture these slower fluctuations in firing rate.

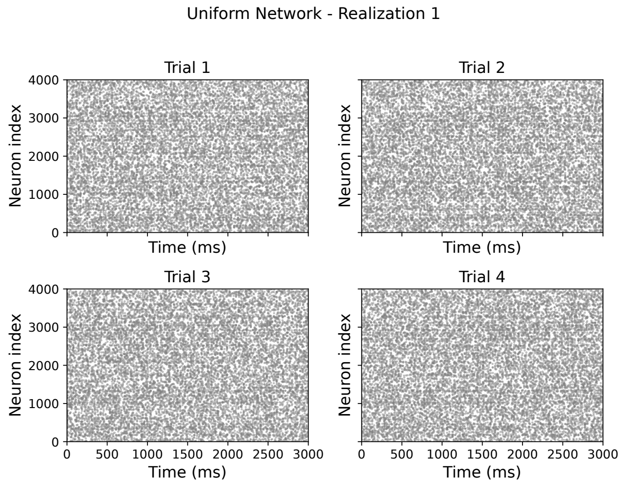
In this study, we find that introducing clustered excitatory connectivity creates slow dynamics. Clusters transition between states of high and low firing, such that neurons exhibit both high spike-time variability and slow firing-rate fluctuations. Introducing stimuli can bias the network toward particular activity states, reducing variability.

2. Results

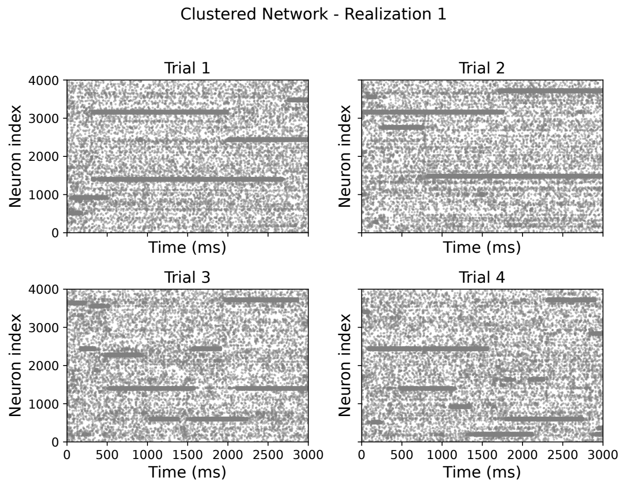
We present our reproduction of the key findings from the original study.

2.1. Metastable Dynamics in Clustered Networks

An excitatory-inhibitory (E/I) network with clustered E→E connectivity produces metastable "high" firing activity states within clusters that are stable yet quick to transition. These high activity states persist for hundreds of milliseconds to seconds before switching, leading to slow rate dynamics and elevated trial-to-trial spike count variability (super-Poisson) compared to unclustered networks.



(a) Uniform (unclustered) network.



(b) Clustered network.

Figure 1. Spike rasters across multiple trials.

2.1.1 Mean Rate

Despite the switching between high and low firing states, the clustered network exhibits a low mean firing rate similar to that of the unstructured network

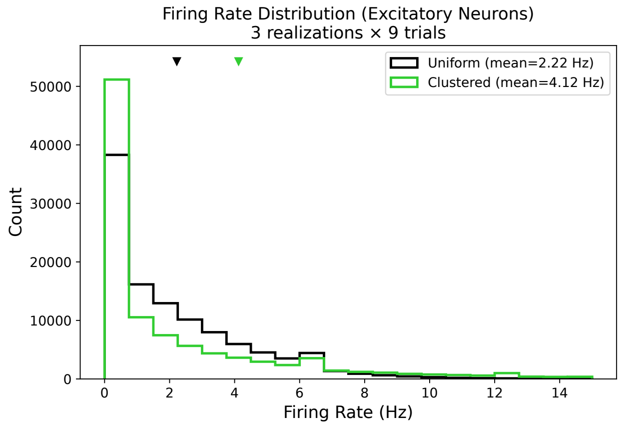


Figure 2. Distribution of firing rates for excitatory neurons in uniform and clustered networks. Both networks maintain low mean firing rates, though the clustered network shows a broader distribution reflecting the alternation between high and low activity states. Triangles indicate mean values.

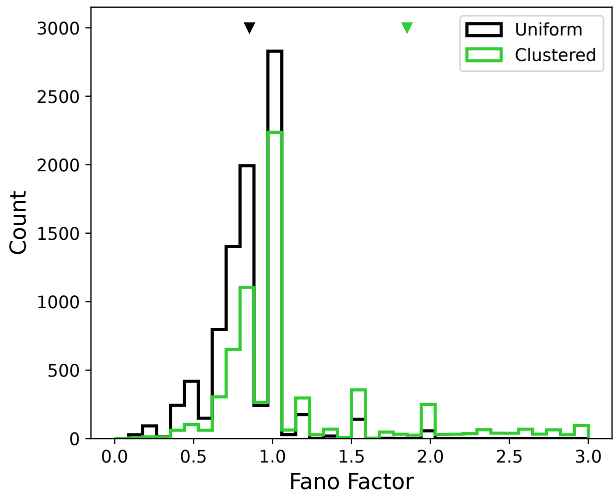


Figure 3. Distribution of Fano factors across neurons for uniform and clustered networks. The clustered network exhibits a broader distribution with a higher mean (triangles), indicating greater spike count variability.

(Figure 2). This is likely due to additional suppression within the network during high firing activity states. The high activity epochs in clustered networks do not alter the overall mean firing rate; however, the firing rate distribution broadens somewhat, likely reflecting the alternation between high and low activity states.

2.1.2 Variability

Neurons in the clustered network exhibit Fano factors greater than one ("super-Poisson" spike count variability), as shown in Figure 3. Furthermore, the Fano factor increases with the size of the counting window (Figure 4). This pattern illustrates the presence of slow firing rate variability in clustered networks—larger windows capture more of the slow rate fluctuations.

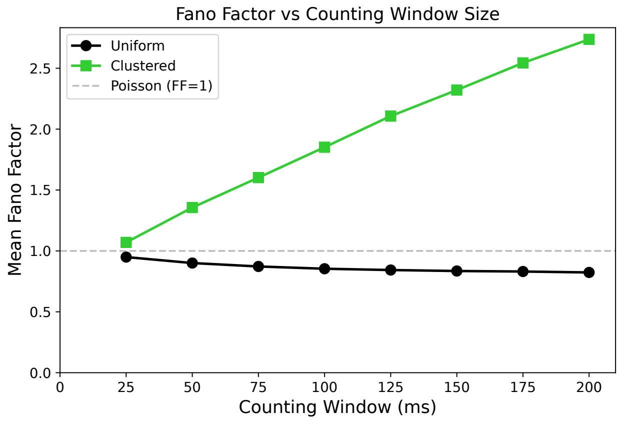


Figure 4. Mean Fano factor as a function of counting window size. In the clustered network, the Fano factor increases with window size, reflecting slow rate fluctuations. The uniform network remains near the Poisson baseline (FF = 1).

2.1.3 Correlations

Pairwise spike count correlations remain near zero on average for both network types (Figure 5). However, in the clustered network, within-cluster neuron pairs show a heavier positive tail in the distribution of correlation coefficients (Figure 6). This observation is consistent with the notion that clusters undergo additional correlated firing rate transitions—switching between high and low activity states—when examined across temporal windows spanning the simulation. Furthermore, this positive tail becomes increasingly pronounced as the counting window size increases (50 ms vs. 100 ms), as expected given that within-cluster neurons share these slow rate fluctuations.

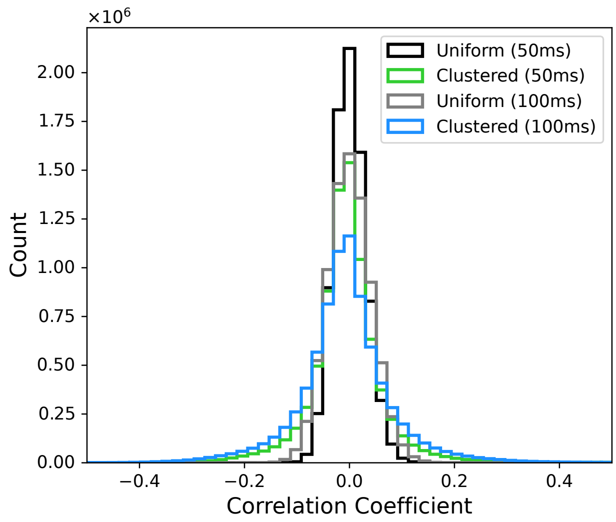


Figure 5. Distribution of pairwise spike count correlations for all neuron pairs. Correlations remain centered near zero on average for both network types.

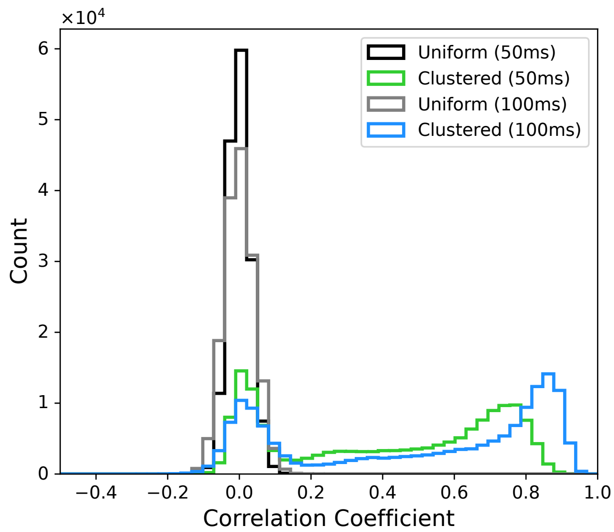


Figure 6. Distribution of pairwise spike count correlations for within-cluster pairs only. The clustered network shows a heavy positive tail that increases with counting window size (50 ms vs. 100 ms).

2.1.4 Timescales

Clustered networks exhibit substantially longer correlation times than unstructured networks. The autocovariance function decays much more slowly as a function of lag in clustered networks compared to their unstructured counterparts. Similarly, the cross-covariance for within-cluster neuron pairs decays slowly, indicating a shared correlated rate change among neurons belonging to the same cluster.

2.2. Critical Cluster Strength

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3. Discussion

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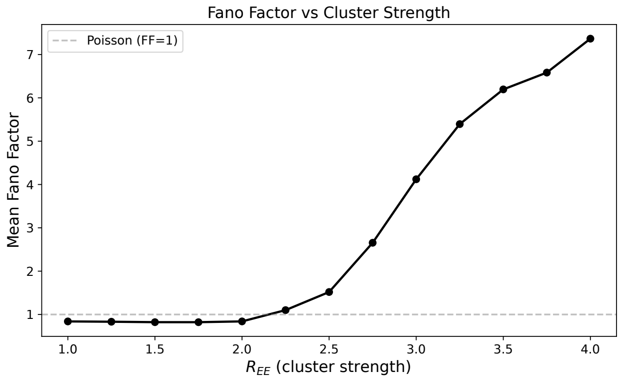

Figure 7. Mean Fano factor as a function of cluster strength (R_{EE}). As within-cluster connection strength increases, the network transitions from Poisson-like variability to super-Poisson variability. The dashed line indicates the Poisson baseline (FF = 1).

4. Methods

4.1. Model

TODO

4.2. Analysis

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References